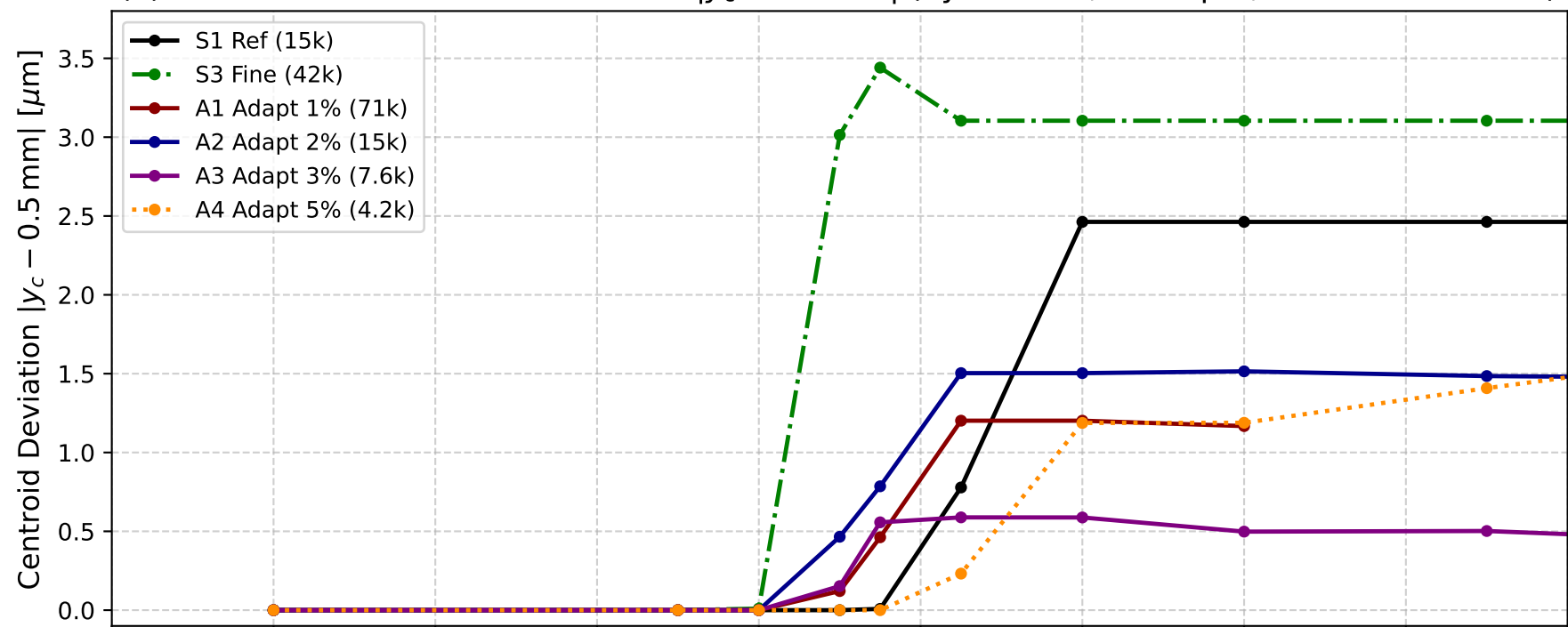


Rigorous Separation of Crack-Centroid Trajectory and Localization Band Width

(a) Crack-Centroid Vertical Deviation $|y_c - 0.5 \text{ mm}|$ (Symmetric, $\leq 3.1 \mu\text{m}$, Element-Bounded)



(b) Diffusive Localization-Band Half-Width ($d \geq 0.50$, $10\text{--}15 \mu\text{m}$, Mesh-Sensitive)

